

Difference between Broadcast and Multicast

The process of delivering data from a source device to a destination device is known as transmission. In a network, data is converted into packets and sent to its destination. There are several methods of transmission, two of which are broadcast and multicast. In summary, multicasting is much more efficient and uses less bandwidth than broadcasting. The transmission technologies broadcast and multicast differ because, in broadcast, the packet is transmitted to all hosts connected to the network. However, the packet is only transmitted to the intended recipients in a multicast.

In this article, you will learn about the difference between **Broadcast** and **Multicast**. But before discussing the differences, you must know about Broadcast and Multicast.

What is broadcast?

Broadcast is a transmission mechanism that allows all network hosts to utilize the same communication channel. All other hosts receive a packet transmitted by a host on the network in broadcast mode. This method is also known as "**all-to-all**". A device on a **Local Area Network (LAN)** can be configured to broadcast messages to other devices on the network.

When a host broadcasts a packet, the intended recipient's address is specified in the packet's address field. Now that the packet has been broadcast, it is received by all other network sites. Each host examines the packet's address field after receiving it. If the packet contains the receiving host's address, it is handled by the receiving host. Otherwise, the packet is disregarded.

The broadcast function can address packets to all hosts on the network. To do this, the host transmitting the packet specifies a specific code in the packet's address field. When a packet with a specific code in the address field is sent, it is received and processed by every host in the network. A common example of broadcasting is a wireless network.

What is Multicast?

Multicasting is the simultaneous transmission of a message or set of packets to several networked devices. Multicasting is secured because only a select group of users receive the packets, and this technique reduces the necessary bandwidth because it distributes a single broadcast among several consumers. The most used protocol for multicasting is called **User Datagram Protocol (UDP)**.

Several applications use multicasting. A user may send them to part of the recipients instead of emails to everyone on the contact list. Multicasting is also utilized in webcasts and video conferencing. Additionally, with the use of multicasting, devices like routers and switches deliver packets to a collection of connected devices.

Nowadays, multicasting has a variety of applications. For example, assessing a distributed database necessitates multicasting. The information in a distributed database is kept in multiple locations. As a result, the user's information request is broadcast to all database locations, and the site has the intended information answers. Similarly, similar information is distributed to several customers in business, as is news distribution, teleconferencing, and distance learning.

Key differences between the Broadcast and Multicast



Here, you will learn the various key differences between **Broadcast** and **Multicast**. Various differences between Broadcast and Multicast are as follows:

1. Broadcasting is a method of sending a message to all recipients simultaneously. In contrast, multicasting is a group communication method in which data is sent simultaneously to a group of target computers.
2. Broadcasting consumes bandwidth since packets are given to sites that are not interested in receiving them. In contrast, multicasting successfully utilizes bandwidth because the packet is transmitted only to hosts that are interested in receiving it.
3. The transmission of a packet in broadcast is one-to-all, whereas the transmission of a packet in multicast is one-to-many.

4. Broadcasting creates a large amount of network traffic by delivering each packet to every site on the network. In contrast, multicasting controls traffic by delivering packets only to interested hosts, lowering network load.
5. In broadcasting, no group administration is necessary. In contrast, group management is needed in multicasting to establish the network in which at least one host is interested in receiving the packet.
6. Broadcast creates a large amount of traffic that can slow down the computer system. However, when compared to broadcasting, multicast generates less traffic, which speeds up the system.
7. Broadcasting is slower. In contrast, multicast is faster.

Head-to-head comparison between the Broadcast and Multicast

Here, you will learn the head-to-head comparisons between Broadcast and Multicast. The main differences between Broadcast and Multicast are as follows:

Features	Broadcast	Multicast
Definition	Broadcasting is a method of sending a message to all recipients simultaneously.	It is a group communication method in which data is sent simultaneously to a group of target computers.
Mapping	It contains one-to-all mapping.	It contains one-to-many mapping.
Bandwidth	The bandwidth of the broadcast is wasted.	The bandwidth of multicast is utilized effectively.
Management	It doesn't need any group management.	It needs group management to specify the group of hosts and stations which will receive packets.
Process	The bandwidth process is slow.	The multicast process is fast.
Traffic	It creates a large amount of network traffic by delivering each packet to every site on the network.	It keeps traffic under control by delivering packets only to interested hosts, lowering the network load.

Conclusion

There are two types of transmission: broadcast and multicast. Broadcasting is the process of distributing a packet to every host connected to the system. It is not worth it if each network has thousands of machines. In contrast, multicasting is helpful because it sends the packet to the hosts who are interested in receiving it.

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